

# Night-Vision Wildlife Species Recognition

## Technical Report and Project Documentation

**Module:** ACM 40960, Projects in Maths Modelling

**Programme:** MSc Data and Computational Science, University College Dublin

**Project:** Number 9, AI Animal Image Recognition Challenge

**Authors:** Srivani Konda (25211398), Navya Sri Mungamuri (25200230)

**Repository:** <https://github.com/ACM40960/wildlife-species-recognition>

## Abstract

Camera traps generate large volumes of imagery that must be reviewed manually before it can be used for ecological analysis. A substantial proportion of captures occur after dark, when the trap switches to an infrared flash and produces a grayscale, low-contrast frame. Classifiers trained on daylight photography perform poorly on such frames.

This project trains a convolutional neural network to identify six mammal species in night-time infrared camera-trap imagery drawn from the Caltech Camera Traps dataset. The central methodological commitment is that evaluation is performed on camera locations withheld entirely from training. Frames captured at a single site share a background, and a classifier can therefore achieve a high score by learning the site rather than the animal. A partition that splits images randomly conceals this effect.

On 233 frames from 25 unseen camera locations the model achieves an accuracy of 0.687 (95% confidence interval 0.597 to 0.780, resampled over camera locations) and a macro one-versus-rest ROC AUC of 0.891, against a six-class chance rate of 0.167. Scored on withheld frames from locations that were seen during training, the same model achieves 0.800. The resulting generalisation gap of 0.113 is reported explicitly, as it is the quantity most likely to be overstated by a less careful experimental design.

Two qualifications are stated at the outset. The reported test set was consulted while selecting pipeline components, so 0.687 is a development estimate rather than a measurement on data never examined. Separately, roughly half the test frames carry a human-drawn bounding box that would not exist at inference time; re-running the entire pipeline with those boxes discarded and a detector substituted yields 0.682, confirming that the headline does not depend on that assistance.

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## 1. Introduction

### 1.1 Background

Motion-triggered camera traps are now standard instrumentation in wildlife monitoring. A single deployment can return tens of thousands of frames per season, and the labour required to sort them is the principal constraint on how much of that data is ever analysed. Automated species classification addresses the constraint directly.

Night captures present a specific difficulty. When ambient light falls below a threshold the trap illuminates the scene with an infrared flash, and the resulting frame is effectively single-channel, low in contrast, and frequently affected by motion blur. Colour and fine texture, which carry much of the discriminative signal in daylight imagery, are largely absent.

### 1.2 Problem statement

Given a night-time infrared camera-trap frame, assign it to one of six species: bobcat, coyote, deer, opossum, rabbit, or raccoon. The task is single-label classification over a fixed, closed set.

### 1.3 Objectives

1. Assemble a reproducible dataset of night infrared frames with verifiable provenance.
2. Train a classifier that performs acceptably on this imagery under transfer learning.
3. Evaluate the classifier in a manner that measures generalisation to new camera sites rather than memorisation of familiar ones.
4. Report the outcome using the metrics appropriate to multi-class classification, with quantified uncertainty.

### 1.4 Scope

The study is deliberately narrow. It concerns six species from one geographic region, evaluated under one random seed and one location partition. Section 9 sets out the consequences of these choices. No claim is made regarding transfer to other regions, other taxa, or other camera hardware.

## 2. Related work

Beery, Van Horn and Perona (2018) introduced the Caltech Camera Traps dataset together with the observation that motivates this project: classifiers evaluated on held-out images from familiar locations substantially overstate performance at new locations. Their paper frames the problem as generalisation to unseen environments, and the location-held-out protocol adopted here follows directly from it.

Norouzzadeh et al. (2018) demonstrated that deep convolutional networks can identify, count and describe animals in camera-trap imagery at scale, establishing the general viability of the approach. Tabak et al. (2019)

reported comparable findings across a different set of North American species. Schneider et al. (2020) examined the factors that determine recognition performance in practice and identified dataset composition and location diversity among the most influential.

For the detection stage, Beery, Morris and Yang developed MegaDetector, a detector trained specifically on camera-trap imagery that distinguishes animal, person and vehicle. Its relevance here is that it retains high recall on infrared frames, which general-purpose detectors trained on consumer photography do not.

The classification backbone is the residual network of He et al. (2016). Calibration follows the temperature scaling method of Guo et al. (2017).

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## 3. Data

### 3.1 Source and licensing

All imagery derives from the Caltech Camera Traps dataset (Beery, Van Horn and Perona, 2018), obtained through the Labeled Information Library of Alexandria: Biology and Conservation (LILA BC) mirror hosted on Google Cloud Storage. The dataset comprises approximately 243,000 camera-trap frames from the American Southwest, annotated with species labels, bounding boxes, capture timestamps and a camera location identifier.

Caltech Camera Traps is released under the CDLA-Permissive licence.

The dataset was selected for two properties. First, it carries camera location identifiers, without which a location-held-out partition cannot be constructed. Second, its night-time frames are genuine infrared captures rather than artificially darkened daylight photography.

### 3.2 Selection criteria

**Infrared confirmation.** A frame was retained only if its capture hour fell between 19:00 and 07:00 and its mean HSV saturation was below 6.0. The saturation test is the operative one; the timestamp filter alone admits colour frames captured at night under artificial illumination.

**Single-species frames only.** Frames annotated with more than one species were excluded, the task being single-label. Annotations for empty frames and vehicles were not treated as a second species.

**Sequence de-duplication.** Camera traps fire in bursts, and consecutive frames within a burst are near-identical. One frame per capture sequence was retained. Without this constraint, near-duplicates would be distributed across the training and test partitions.

**Stratification across sites and time.** Candidates were drawn round-robin across camera locations, and within each location at evenly spaced intervals across that location's date range, subject to a cap of 20 frames per location. This prevents a small number of heavily-trafficked sites from dominating a species.

**Determinism.** The candidate ordering is fixed before any download begins, and both retention and filename assignment follow that ordering rather than the order in which concurrent downloads complete. Repeating the build with the same seed reproduces the dataset byte for byte.

The resulting dataset comprises 1,200 frames: 200 per species, distributed across 117 camera locations.

### 3.3 Storage convention

Frames are stored uncropped, downscaled so that the longer edge measures 384 pixels. The animal bounding box is recorded in the manifest, and the crop is applied when the image is loaded rather than being written into the file.

This convention proved consequential in practice. The crop strategy was revised three times during the project. Had crops been written into the stored files, each revision would have required the dataset to be rebuilt from source.

### 3.4 Annotation provenance

Caltech Camera Traps supplies ground-truth bounding boxes for approximately half of the selected frames. Coverage was extended by running detectors over the remainder. The origin of every box is recorded in the manifest, so that the reported coverage figure is auditable rather than asserted.

| Provenance                              | Frames       | Proportion of dataset |
|-----------------------------------------|--------------|-----------------------|
| Ground truth, supplied with the dataset | 598          | 49.8%                 |
| MegaDetector                            | 228          | 19.0%                 |
| YOLOv8                                  | 200          | 16.7%                 |
| No box detected                         | 174          | 14.5%                 |
| <b>Total with a bounding box</b>        | <b>1,026</b> | <b>85.5%</b>          |

Recording provenance is not bookkeeping for its own sake. A ground-truth box is an oracle that does not exist at inference time, so any score computed partly on such crops overstates what the deployed pipeline would achieve. Without this column that contamination is invisible. Section 6.4 uses it to separate the two cases and to quantify the difference. An earlier defect in the box-fill routine, which silently relabelled detector-derived boxes as ground truth on a second run, was detectable only because the column exists; the regression test `test_box_fill_preserves_provenance_on_rerun` in `tests/test_data_config.py` guards against it.

The 174 frames without a box are retained and classified from the full frame after letterbox padding.

### 3.5 Partitioning

The module `src/split.py` assigns each camera location in its entirety to a single partition.

| Partition  | Frames | Locations |
|------------|--------|-----------|
| Training   | 735    | 69        |
| Validation | 232    | 23        |
| Test       | 233    | 25        |

The assignment procedure is greedy and class-aware. It fills the validation and test targets on a per-species basis so that every species is represented in every partition, and it declines to assign a location if doing so would leave any species without a training location. The procedure is deterministic given the seed.

A second held-out set is carved from the training locations. Fifteen per cent of training-location frames are withheld from training, selected by a stable hash of the image identifier so that membership is invariant across runs and processes. These 110 frames constitute the seen-location test set. Scoring the same model on both sets is what renders the generalisation gap in Section 6.3 measurable.

## 4. Methodology

### 4.1 Preprocessing

1. Crop to the recorded animal bounding box, with padding of 15 per cent of the box dimensions on each side.
2. Convert to grayscale and replicate across three channels, matching the input specification of the ImageNet-pretrained backbone.
3. Letterbox to 224 by 224 pixels: resize preserving aspect ratio, then pad to square with a neutral grey.
4. Normalise using ImageNet channel statistics.

Step 3 replaced an earlier centre-crop. Centre-cropping a 4:3 frame discards roughly 35 per cent of its width and was observed to remove animals positioned near the frame edge.

Training applies additional augmentation: random resized crop at scale 0.8 to 1.0, horizontal flip, rotation within 10 degrees, brightness and contrast jitter, and augmentation specific to infrared imagery consisting of gamma jitter, sharpness jitter and random erasing at probability 0.25. Validation and test transforms are deterministic.

Training, evaluation and single-image inference all invoke the same cropping function. This is a deliberate constraint. An earlier revision applied cropping during training but classified full frames at inference, and the resulting discrepancy caused a substantial loss of accuracy at serving time before it was identified. The measurement predates the current experiment log and is not recorded there, so no figure is quoted; the regression test `test_predict_uses_same_crop_as_training` in `tests/test_data_config.py` exists to prevent a recurrence.

## 4.2 Detection stage

Two detectors are available, both optional, and both used to populate bounding boxes in the manifest.

MegaDetector version 5a is trained on camera-trap imagery including infrared captures and predicts three classes. A COCO-pretrained YOLOv8n model is available as an alternative.

The class label assigned by either detector is discarded; only the box is retained. The relevant performance characteristic is therefore recall on grayscale infrared frames. The COCO-pretrained model raised coverage from 50 per cent to 66 per cent with no measurable effect on accuracy. MegaDetector raised coverage to 86 per cent and contributed approximately 0.07 accuracy.

Every detector in the project returns boxes as `(x, y, w, h)` in original image pixels, matching the manifest encoding. A detector returning corner coordinates would produce an incorrect crop without raising an error.

## 4.3 Model architecture

The classifier is a ResNet-18 initialised from ImageNet weights, with the final fully-connected layer replaced by a new six-way linear layer.

| Parameter      | Setting                                 | Rationale                                              |
|----------------|-----------------------------------------|--------------------------------------------------------|
| Backbone       | ResNet-18                               | Trainable on CPU within the project timeframe          |
| Initialisation | ImageNet pretrained                     | 1,200 images is insufficient for training from scratch |
| Frozen         | <code>conv1, bn1, layer1, layer2</code> | Early layers encode generic edge and texture features  |
| Retrained      | <code>layer3, layer4, fc</code>         | Semantic layers require adaptation to infrared imagery |

Frozen batch-normalisation layers are additionally placed in evaluation mode during training. Otherwise their running statistics continue to update despite the weights being frozen, which alters the nominally frozen representation over the course of training.

The contribution of transfer learning was measured. Trained from scratch on identical data the model reached 0.62 against 0.77 with pretrained initialisation. Both figures were obtained under the earlier random partition and are superseded as absolute values; the 15-point difference is the relevant quantity.

## 4.4 Training procedure

| Parameter     | Value                                                                       |
|---------------|-----------------------------------------------------------------------------|
| Optimiser     | AdamW                                                                       |
| Learning rate | 3e-4, cosine annealing schedule                                             |
| Weight decay  | 1e-4                                                                        |
| Loss          | Cross-entropy with inverse-frequency class weights and label smoothing 0.05 |
| Batch size    | 32                                                                          |
| Epochs        | 16, early stopping at patience 5                                            |
| Random seed   | 42, deterministic algorithms enabled                                        |

The reported run terminated at epoch 15 under early stopping, with the best validation accuracy of 0.728 recorded at epoch 10. Only the best-validation checkpoint is retained.

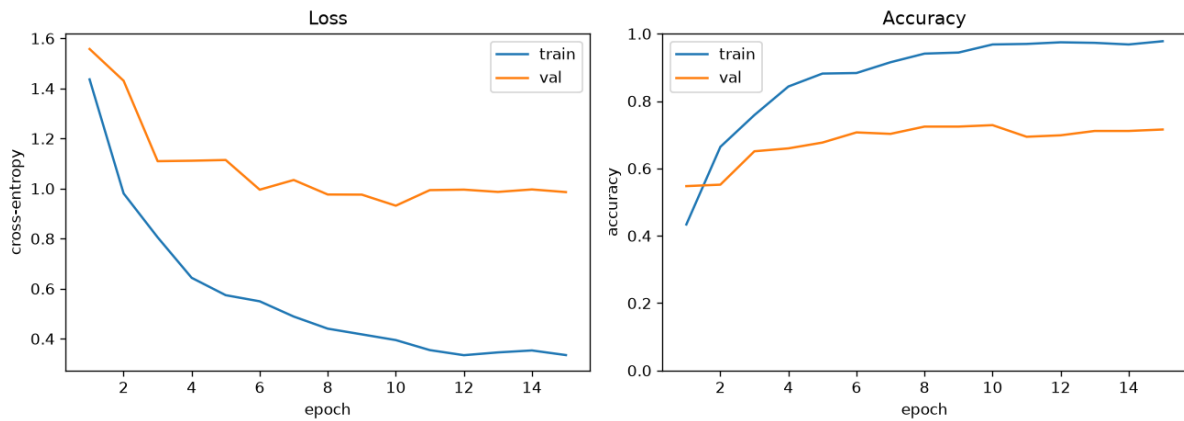


Figure 1. Training and validation loss and accuracy per epoch. The widening gap between the training and validation curves after roughly epoch 8 is the point at which the model begins fitting the training locations rather than the species, and is what early stopping is guarding against.

## 5. Evaluation framework

Accuracy alone is an inadequate summary of a classifier. The confusion matrix is the appropriate starting point, and precision, recall, F1 and AUC-ROC follow from it. This section is ordered accordingly.

### 5.1 Classification outcomes

Relative to any given species, each prediction falls into one of four categories:

- True positive: the frame depicts the species and the model predicted it.
- False positive: the frame depicts another species and the model predicted this one.
- True negative: the frame depicts another species and the model predicted another species.
- False negative: the frame depicts the species and the model predicted otherwise.

For six classes these quantities are computed one species at a time under a one-versus-rest scheme and then averaged.

### 5.2 Confusion matrix

Rows correspond to true species and columns to predicted species, so correct classifications occupy the diagonal.

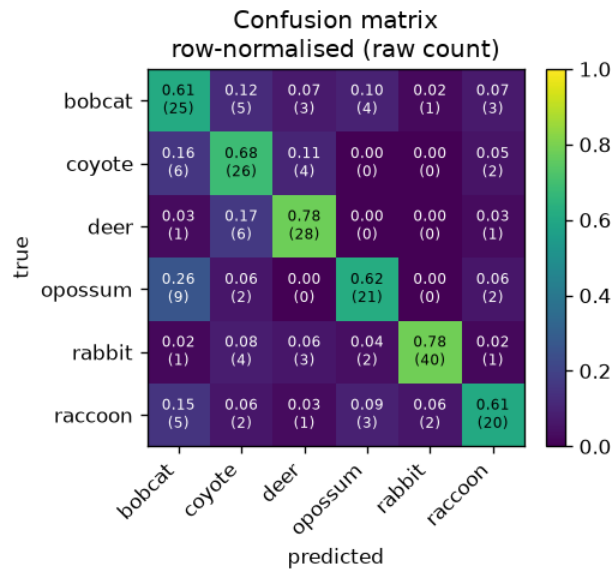


Figure 2. Confusion matrix on the 233 unseen-location test frames. Each cell shows the row-normalised proportion with the raw count beneath it.

Regarding normalisation, the general recommendation is to normalise rows to unit sum, with an exception where class imbalance is itself informative. The matrix presented here does both. Row normalisation renders the six rows comparable despite unequal test supports, ranging from 33 to 51 frames, while the counts preserve visibility of that imbalance.

The dominant error mode is bidirectional confusion between bobcat and coyote. This is not arbitrary. The two are quadrupedal carnivores of comparable size, and the coat patterning that distinguishes them is substantially absent from a grayscale infrared frame.

### 5.3 Metrics reported

| Metric                           | Definition                                                  | Purpose                                       |
|----------------------------------|-------------------------------------------------------------|-----------------------------------------------|
| Accuracy                         | Correct predictions divided by total predictions            | Headline summary, insufficient alone          |
| Balanced accuracy                | Mean of per-class recall                                    | Insensitive to unequal test supports          |
| Precision (macro)                | Of frames predicted as a species, the proportion correct    | Quantifies the cost of false alarms           |
| Recall (macro)                   | Of frames depicting a species, the proportion identified    | Quantifies the cost of misses                 |
| F1 (macro)                       | Harmonic mean of precision and recall                       | Single summary where both matter equally      |
| ROC AUC (macro, one-versus-rest) | Area under recall against false-positive rate               | Threshold-independent ranking quality         |
| Top-2 and top-3 accuracy         | True class within the leading 2 or 3 predictions            | Relevant to human-in-the-loop review          |
| Expected calibration error       | Discrepancy between stated confidence and observed accuracy | Whether a reported probability is trustworthy |

All averages are macro averages, so each species contributes equally irrespective of its test support.

The choice of primary metric depends on application. The intended use is triage of a night's captures prior to human review, in which a missed species is more costly than a false alarm subsequently rejected by a reviewer. Recall is therefore the operative metric, and balanced accuracy is macro recall by construction.

## 5.4 ROC and AUC

Receiver operating characteristic curves plot recall against false-positive rate as the decision threshold varies. The diagonal represents random assignment. The macro AUC is 0.891.

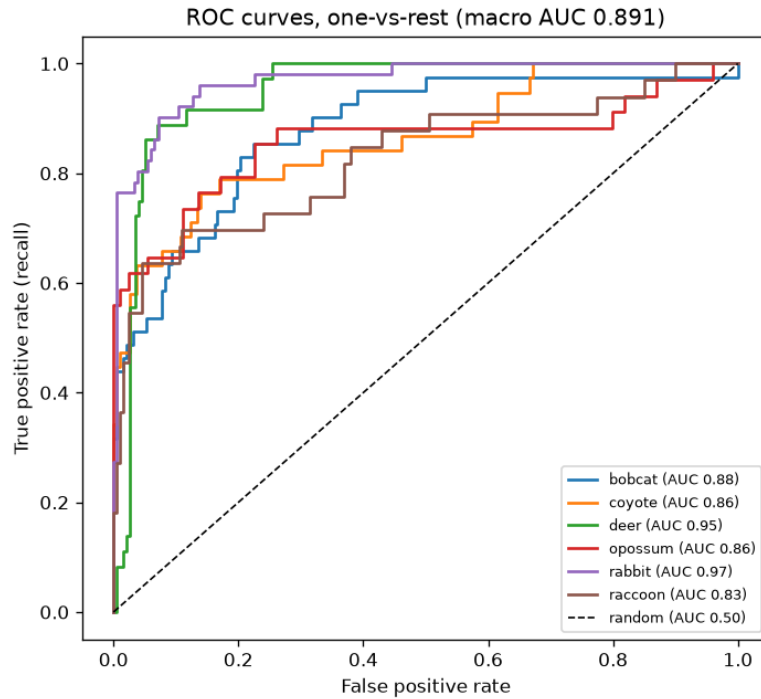


Figure 3. One-versus-rest ROC curves, one per species, with per-class AUC in the legend.

The separation between an AUC of 0.891 and an accuracy of 0.687 is informative. It indicates that when the leading prediction is incorrect, the correct species is nonetheless ranked highly. The top-2 accuracy of 0.828 corroborates this. A model exhibiting accuracy of 0.687 alongside an AUC near 0.70 would be discriminating poorly; the present model discriminates well and is then required to resolve a genuinely difficult binary distinction.

## 5.5 Interval estimation

The test set comprises 233 frames drawn from 25 camera locations. Point estimates alone would overstate the precision of the result.

Intervals are computed by a cluster bootstrap that resamples whole camera locations with replacement, taking every frame belonging to a drawn location. This is deliberate. Frames from one camera share a background, a season and frequently the same individual animal, so they are not independent observations. The partition is location-grouped for exactly that reason, and an interval computed over individual frames would contradict the unit of generalisation being claimed.

The difference is material. Resampling frames gives 0.625 to 0.743; resampling locations gives 0.597 to 0.780, an interval roughly 55 per cent wider. The clustered interval is the one quoted throughout this report. Both are recorded in the output, with the frame-level version marked as naive.

## 5.6 Calibration

A softmax output of 0.9 should correspond to a correct prediction approximately nine times in ten. Expected calibration error quantifies the discrepancy by binning predictions by confidence and comparing mean confidence to observed accuracy within each bin.

Temperature scaling (Guo et al., 2017) is applied. A single scalar is fitted by minimising negative log likelihood on the validation partition, never on the test partition. Division of the logits by this scalar cannot alter their ordering and therefore cannot alter accuracy. The fitted value was 1.07, and expected calibration error improved from



0.054 to 0.048.

The larger part of the calibration improvement over the course of the project, from 0.150 to 0.054, is attributable to improvements in the model rather than to the calibration step.

## 6. Results

All figures below derive from a single run evaluated on the 233-frame unseen-location test set.

### 6.1 Aggregate performance

| Metric                           | Value | 95% confidence interval |
|----------------------------------|-------|-------------------------|
| Accuracy                         | 0.687 | 0.597 to 0.780          |
| Balanced accuracy                | 0.680 |                         |
| Precision (macro)                | 0.691 |                         |
| Recall (macro)                   | 0.680 |                         |
| F1 (macro)                       | 0.682 | 0.595 to 0.762          |
| ROC AUC (macro, one-versus-rest) | 0.891 |                         |
| Top-2 accuracy                   | 0.828 |                         |
| Top-3 accuracy                   | 0.893 |                         |
| Expected calibration error       | 0.048 | at temperature 1.07     |
| Mean confidence                  | 0.691 |                         |

Mean confidence of 0.691 against accuracy of 0.687 indicates that the model's stated certainty is well matched to its performance.

### 6.2 Per-species performance

| Species | Precision | Recall | F1    | Test frames |
|---------|-----------|--------|-------|-------------|
| Rabbit  | 0.930     | 0.784  | 0.851 | 51          |
| Deer    | 0.718     | 0.778  | 0.747 | 36          |
| Opossum | 0.700     | 0.618  | 0.656 | 34          |
| Raccoon | 0.690     | 0.606  | 0.645 | 33          |
| Coyote  | 0.578     | 0.684  | 0.627 | 38          |
| Bobcat  | 0.532     | 0.610  | 0.568 | 41          |

Rabbit and deer are the most reliably identified, consistent with their being the most distinctive in silhouette. Bobcat and coyote are the least reliable and, as the confusion matrix shows, are principally confused with each other.

Test supports vary from 33 to 51 despite the dataset being exactly balanced at 200 frames per species. This follows necessarily from partitioning by location, since camera sites do not contain equal numbers of each species. The imbalance is a direct cost of the partitioning protocol and is the reason balanced accuracy is reported alongside accuracy.

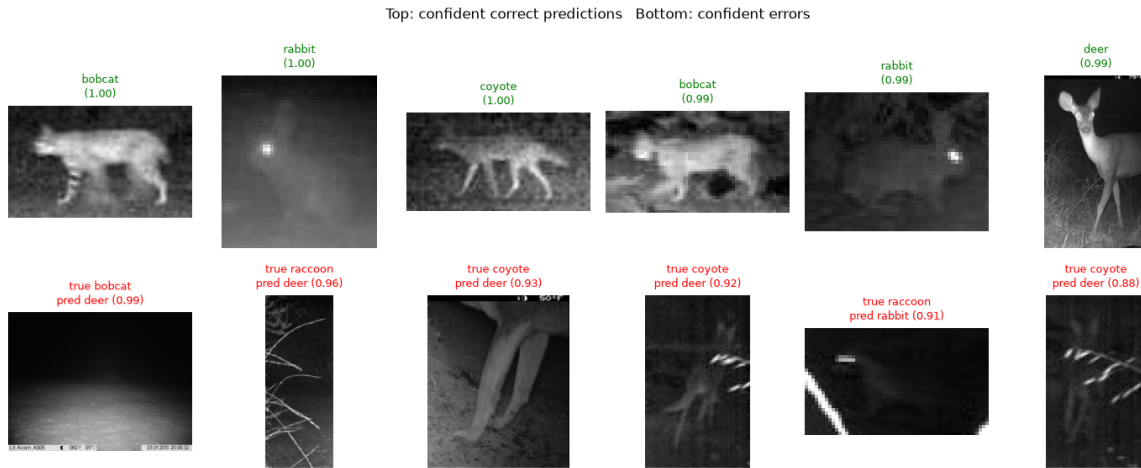


Figure 4. Representative predictions. The upper row shows the most confident correct classifications; the lower row shows the most confident errors. Confident mistakes cluster around dark, occluded or partially framed animals and around the bobcat and coyote pair.

### 6.3 Generalisation to unseen locations

The same model, evaluated on both held-out sets:

| Camera locations                      | Frames | Accuracy     |
|---------------------------------------|--------|--------------|
| Seen during training, frames withheld | 110    | 0.800        |
| Never seen during training            | 233    | 0.687        |
| <b>Difference</b>                     |        | <b>0.113</b> |

The 11-point reduction when the camera is unfamiliar is the principal finding of this study. A random partition would have reported a figure close to 0.800 and would have concealed the effect entirely.

The gap measured 0.17 earlier in the project. Cropping to the animal and substituting letterbox padding for centre-cropping reduced it to 0.113, which is the expected direction: the less background the model observes, the less of the site it can learn.

### 6.4 Dependence on ground-truth boxes

Roughly half the test frames carry a bounding box drawn by the compilers of Caltech Camera Traps. Such a box is an oracle: at inference on a new frame from a new camera, no one supplies the animal's location. A score computed partly on those crops is therefore not a measurement of the deployable pipeline.

Partitioned by whether a ground-truth box was available, the difference is substantial:

| Test frames                | Count | Accuracy |
|----------------------------|-------|----------|
| Ground-truth box available | 108   | 0.778    |
| No ground-truth box        | 125   | 0.608    |

The entire test set was therefore re-evaluated with all ground-truth boxes discarded and MegaDetector run over every frame, which reproduces the conditions of deployment exactly.

| Configuration                                   | Accuracy     | AUC          | Box coverage |
|-------------------------------------------------|--------------|--------------|--------------|
| Manifest boxes, approximately half ground truth | 0.687        | 0.891        | 84%          |
| <b>Detector only, end to end</b>                | <b>0.682</b> | <b>0.900</b> | 74%          |

The end-to-end result of 0.682 falls within the confidence interval of the reported 0.687. Removing the oracle reduces box coverage from 84 per cent to 74 per cent, yet accuracy is essentially unchanged and AUC improves marginally. The headline figure does not depend on the ground-truth annotations.

What does affect performance is whether a frame receives a box at all. Within the detector-only run, frames in which an animal was located score 0.808; frames in which nothing was located score 0.328. Detection coverage, rather than classifier capacity, is the operative constraint on further improvement.

## 6.5 Ablation

Changes were introduced individually.

| Configuration                                            | Unseen-location accuracy |
|----------------------------------------------------------|--------------------------|
| Baseline, version 3 pipeline with detected boxes         | 0.549                    |
| With letterbox padding and infrared augmentation         | 0.614                    |
| With MegaDetector boxes, coverage raised from 66% to 86% | 0.687                    |

For comparison, on the identical partition and the same version 3 pipeline, classifying the full frame rather than the animal crop yields 0.459 with an expected calibration error of 0.22, against 0.549 for the cropped baseline in the table above. The isolated effect of cropping is therefore approximately 9 percentage points of accuracy, together with a reduction in calibration error of roughly three quarters.

The wider gap between full-frame classification at 0.459 and the final pipeline at 0.687, approximately 23 percentage points, should not be attributed to cropping alone. It is the cumulative effect of cropping together with letterbox padding, infrared augmentation and the improved box coverage from MegaDetector.

## 6.6 Negative results

Two interventions did not produce the expected benefit. Both are recorded deliberately. An experimental log that contains only successes cannot be used to judge whether a reported gain is real or selected after the fact, and a reader has no way to distinguish a pipeline that was reasoned through from one that was tuned until the number improved. Recording what failed also prevents the same avenue being explored again, and in one case documented below it corrected a claim this project had already made in error.

**Test-time augmentation degraded performance.** Averaging predictions over the frame and its horizontal mirror is ordinarily a low-cost improvement. Here it yielded 0.682 against 0.687, with expected calibration error worsening from 0.048 to 0.072, at twice the inference cost. It is disabled by default. The result had been provisionally written up as an improvement before it was measured, which is precisely the error the log exists to prevent.

**Temperature scaling did not affect accuracy**, and could not have done. It is a monotonic rescaling of the logits and cannot alter the arg max. Its effect is confined to calibration, which is its only claim.

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## 7. Reproducibility

### 7.1 Recorded artefacts

Each run writes the following alongside its checkpoint:

| File                                                    | Contents                                                                                |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>config.json</code>                                | Complete hyperparameter configuration                                                   |
| <code>environment.json</code>                           | Versions of Python, PyTorch, torchvision, NumPy and scikit-learn, and the device used   |
| <code>history.csv</code> ,<br><code>history.json</code> | Per-epoch training and validation loss and accuracy, learning rate, epoch duration      |
| <code>metrics.json</code>                               | All quantities reported in Section 6                                                    |
| <code>class_names.txt</code>                            | Class ordering, preventing a checkpoint from being interpreted against incorrect labels |
| <code>best_model.pt</code>                              | Best-validation weights together with class names and configuration                     |

These artefacts exist so that a reported number can be traced to the exact conditions that produced it. A figure quoted without its configuration cannot be checked, defended under questioning, or compared against a later run, and after several weeks of experimentation no one reliably remembers which settings produced which result. Writing the environment alongside the configuration additionally distinguishes a genuine change in behaviour from a change in library versions. The class ordering is recorded because a checkpoint loaded against a differently ordered label set produces confident and entirely wrong predictions with no error raised, which is among the harder failures to detect after the fact.

## 7.2 Determinism

Random seeds are set for Python, NumPy and PyTorch. cuDNN autotuning is disabled and deterministic algorithms are requested. The dataset build, the location partition and the seen-location carve are each seeded and order-stable.

The reported run was reproduced from scratch on a rebuilt machine and returned identical figures: best validation accuracy 0.728 at epoch 10, test accuracy 0.687, fitted temperature 1.070, seen-location accuracy 0.800.

## 7.3 Software environment

An environment specification is provided so that a collaborator may reproduce the environment without installing packages individually:

```
conda env create -f environment.yml
conda activate night-wildlife
```

The two detector packages are commented out in the environment file, as they are required only to regenerate bounding boxes; the boxes themselves are committed to the manifest.

The reported result was obtained under the following versions:

| Component    | Version |
|--------------|---------|
| Python       | 3.11.15 |
| PyTorch      | 2.13.0  |
| torchvision  | 0.28.0  |
| NumPy        | 2.3.5   |
| scikit-learn | 1.9.0   |
| Device       | CPU     |

# 8. Repository organisation

```

src/
  config.py          Configuration dataclass, single source of settings
  data.py            Loading, transforms, load-time cropping, partitioning
  split.py           Location-grouped partition
  model.py           Backbone construction, freezing, batch-norm handling
  train.py           Training loop, early stopping, checkpointing
  evaluate.py        Metrics, intervals, calibration, error analysis
  detect.py          YOLOv8 wrapper, optional
  megadetector.py    MegaDetector wrapper, optional
  utils.py           Seeding, determinism, figure generation

scripts/
  build_night_wildlife.py Dataset construction from source
  validate_dataset.py    Integrity checks
  fill_boxes_yolo.py     Bounding-box population
  make_detector_manifest.py Detector-only manifest for end-to-end evaluation
  run_training.py         Training entry point
  run_evaluation.py       Evaluation entry point
  predict.py             Single-image inference

docs/
  METHODOLOGY.md        Detailed pipeline description
  experiments.md         Run log
  literature_review.md   Background reading
  DATA_LICENSE.md       Dataset licensing and attribution
  demo_results/          Committed figures and metrics for the reported run

data/night_wildlife/    1,200 frames and manifest.csv
tests/                  43 tests

```

## 9. Limitations

**Scale and scope.** Six species and 1,200 frames. The reported accuracy applies to these species at these unseen Caltech Camera Traps sites. No claim of transfer to other regions, taxa or camera hardware is made.

**The test set is not untouched.** Crop strategies, detector choice, augmentation and test-time augmentation were each compared by inspecting unseen-location test accuracy. The test set therefore influenced pipeline selection, and 0.687 is a development estimate that is likely to be slightly optimistic. Correct practice would select on validation data or on nested location-grouped cross-validation and open the test set once. Repairing this requires camera locations untouched by any prior experiment, and none remain in this dataset. Every run is logged so that the extent of the reuse is visible.

**Single partition and single seed.** The study did not cross-validate across multiple location partitions. The reported intervals capture sampling variation across camera locations but not variation arising from which locations were assigned to the test set in the first place.

**Incomplete detection.** Under the end-to-end configuration, 26 per cent of test frames receive no bounding box and are classified from the full frame. These frames score 0.328 against 0.808 for frames in which an animal is located, and they are the principal drag on the aggregate figure.

**Confusable species.** Bobcat and coyote are not reliably separable by this model at this resolution, with recall of 0.610 and 0.684 respectively. Predictions for these two classes should be treated as provisional.

**Domain restriction.** The model has been trained exclusively on night infrared frames and should not be applied to daylight colour imagery.

**Closed-set assumption.** The classifier has no facility to abstain. A species outside the six will be assigned to one of them, with confidence.

## 10. Conclusions and further work

A ResNet-18 fine-tuned on 1,200 night infrared camera-trap frames identifies six species at 0.687 accuracy and 0.891 macro AUC on camera locations withheld from training. The same model reaches 0.800 on locations seen during training, and the resulting gap of 0.113 quantifies the extent to which a conventional random partition would have overstated the result.

Two design decisions account for most of the measured performance. Cropping to the detected animal is worth approximately 9 percentage points in isolation, measured against a matched full-frame baseline on the same pipeline, and reduces calibration error by roughly three quarters. Substituting letterbox padding for centre-cropping, together with infrared-specific augmentation, contributes a further 0.069. The full distance from the original full-frame baseline of 0.459 to the final 0.687 is approximately 23 percentage points, but that figure is the cumulative effect of every change in the ablation chain rather than of cropping alone.

The end-to-end evaluation in Section 6.4 establishes that the result survives removal of the ground-truth boxes, and in doing so identifies detection coverage as the binding constraint. Frames in which the detector locates an animal are classified at 0.808; frames in which it does not fall to 0.328.

Three directions follow. First, raising detection coverage, whether by lowering the detector confidence threshold or by falling back to a second detector when the first returns nothing, addresses the largest identified source of error. Any such tuning must be selected on validation data, since selecting it on the test set would compound the limitation recorded in Section 9. Second, cross-validation across several independent location partitions would characterise the uncertainty arising from partition composition. Third, aggregation of predictions across a capture sequence using the stored sequence identifiers would change the unit of classification from the frame to the burst and would need to be reported as a separate task.

---

## 11. Statement of contributions

| Author              | Student number | Responsibility                                                               |
|---------------------|----------------|------------------------------------------------------------------------------|
| Srivani Konda       | 25211398       | Data pipeline: dataset construction, preprocessing, partitioning, validation |
| Navya Sri Mungamuri | 25200230       | Model, training procedure, evaluation, reported results                      |

The division is reflected in the commit history, each author committing under their own identity.

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## 12. Licensing and attribution

Source code is released under the MIT Licence.

Imagery derives from Caltech Camera Traps, obtained through LILA BC under the CDLA-Permissive Licence. The 1,200 frames redistributed in the repository are downscaled, uncropped derivatives of the originals.

MegaDetector is the work of Beery, Morris and Yang and is released under the MIT Licence.

YOLOv8n (Ultralytics), used as an optional detector, is licensed under AGPL-3.0. Review its terms before building on the detection stage. It supplied 200 of the dataset's bounding boxes, 16.7% of the manifest.

All figures in this report were generated by the authors from their own experimental runs.

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## Appendix A: Figure inventory

| Figure                                                 | Source file                            | Section |
|--------------------------------------------------------|----------------------------------------|---------|
| Training and validation curves                         | docs/demo_results/training_curves.png  | 4.4     |
| Confusion matrix, row-normalised with raw counts       | docs/demo_results/confusion_matrix.png | 5.2     |
| ROC curves with per-class AUC                          | docs/demo_results/roc_curves.png       | 5.4     |
| Confident correct predictions against confident errors | docs/demo_results/error_analysis.png   | 6.2     |

## Appendix B: Reproduction commands

Validation of the dataset prior to training:

```
python scripts/validate_dataset.py --data-dir data/night_wildlife
```

The reported result:

```
python scripts/run_training.py --data-dir data/night_wildlife --epochs 16 \
  --image-size 224 --pretrained --grayscale --freeze-until layer2 \
  --learning-rate 3e-4 --output-dir results/v5_final --device cpu
python scripts/run_evaluation.py --output-dir results/v5_final --device cpu
```

The end-to-end, detector-only evaluation reported in Section 6.4:

```
python scripts/make_detector_manifest.py --splits test
python scripts/run_evaluation.py --output-dir results/v5_final \
    --manifest-name manifest_detector.csv --device cpu
```

The same-location comparison, for contrast:

```
python scripts/run_training.py --data-dir data/night_wildlife --epochs 16 \
    --image-size 224 --pretrained --grayscale --freeze-until layer2 \
    --learning-rate 3e-4 --split-by stratified \
    --output-dir results/samelocation --device cpu
python scripts/run_evaluation.py --output-dir results/samelocation --device cpu
```

Single-image inference:

```
python scripts/predict.py data/night_wildlife/bobcat/bobcat_0032.jpg \
    --checkpoint results/v5_final/best_model.pt
```